

# Pulkit Verma

✉ verma.pulkit@asu.edu | 🏠 pulkitverma.net | 📧 pulkitverma25 | 🌐 pulkitverma25

## Research Interests

AI Safety, AI Planning, Action-Model Learning, Analysis of Abstractions, and Robotics.

## Education

### Ph.D. in Computer Science, Arizona State University

Advisor: Prof. Siddharth Srivastava | GPA: 4.0/4.0

Tempe, USA

Exp. Fall 2023

### Master of Technology in Computer Science & Engineering, IIT Guwahati

Advisor: Prof. P. K. Das | Thesis: Resource Usage Analysis for Speech Recognition Techniques | GPA: 9.07/10

Guwahati, India

May 2015

### Bachelor of Engineering in Information Technology, SVITS, Indore

Advisor: Prof. Anand Rajawat | Capstone: Smart Analyzer - Predict and Cache Webpages based on Usage.

Indore, India

June 2011

## Publications

### Journals and Conferences

**Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. "Discovering User-Interpretable Capabilities of Black-Box Planning Agents". In *19th International Conference on Principles of Knowledge Representation and Reasoning (KR)*, 2022. 📄

Rashmeet Kaur Nayyar\*, **Pulkit Verma\***, and Siddharth Srivastava. "Differential Assessment of Black-Box AI Agents". In *36th AAAI Conference on Artificial Intelligence*, 2022. 📄 🎥 \*Equal Contribution

Yizhong Wang et al. "Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ Tasks". In *2022 Conference on Empirical Methods in Natural Language Processing*, 2022. 📄

**Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. "Asking the Right Questions: Learning Interpretable Action Models Through Query Answering". In *35th AAAI Conference on Artificial Intelligence*, 2021. 📄 🎥

**Pulkit Verma**, and Pradip K. Das. "A Comparative Study of Resource Usage for Speaker Recognition Techniques". In *International Conference on Signal Processing and Communication*, 2016. 📄

**Pulkit Verma**, and Pradip K. Das. "i-Vectors in Speech Processing Applications: A Survey". In *International Journal of Speech Technology*, 18(4), pp.529-546, 2015. 📄

Mayank Gupta, **Pulkit Verma**, Tuhin Bhattacharya, and Pradip K. Das. "A Mobile Agents based Distributed Speech Recognition Engine for Controlling Multiple Robots". In *International Conference on Advances In Robotics*, 2015. 📄

**Pulkit Verma**, Mayank Gupta, Tuhin Bhattacharya, and Pradip K. Das. "Improving Services using Mobile Agents-based IoT in a Smart City". In *2014 International Conference on Contemporary Computing and Informatics*, 2014. 📄

### Workshops, Demos, Posters, Preprints, DCs

Naman Shah\*, **Pulkit Verma\***, Trevor Angle, and Siddharth Srivastava. "JEDAI: A System for Skill-Aligned Explainable Robot Planning". In *21st International Conference on Autonomous Agents and MultiAgent Systems (Demonstration Track)*, 2022. [Best Demo Award]. 📄 🎥 \*Equal Contribution

**Pulkit Verma**. "Data Efficient Paradigms for Personalized Assessment of Taskable AI Systems". In *ICAPS 2022 Doctoral Consortium*, and *KR 2022 Doctoral Consortium*, 2022. 📄

**Pulkit Verma**. "Data Efficient Algorithms and Interpretability Requirements for Personalized Assessment of Taskable AI Systems". In *IJCAI 2021 Doctoral Consortium*, 2021. 📄

**Pulkit Verma**, and Siddharth Srivastava. "Learning Causal Models of Autonomous Agents using Interventions". In *IJCAI 2021 Workshop on Generalization in Planning*, 2021. 📄

Alok Shankar Mysore et al. "Investigating the 'Wisdom of Crowds' at Scale.". In *Adjunct Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology (Poster)*, 2015. 📄

## Professional Experience

---

### Arizona State University

#### Graduate Research Associate

Tempe, USA

Aug. 2018 - (present)

- Investigating the minimal set of requirements in an AI system that would enable a user to assess and understand the limits of its safe operability.
- Developed a framework which generates interrogation policies to derive an interpretable model of an AI agent.
- Co-lead a team to create an educational tool “JEDAI” to introduce AI planning concepts to laypersons using mobile manipulator robots.
- Co-designed the experiments showcasing possible use-cases of an NSF project on training lay people to use adaptive AI systems which are safe, robust and reliable.

### NetApp Inc.

#### Storage Efficiency Developer

Bengaluru, India

Jul. 2015 - Jul. 2018

- *Workload Prediction*: Predicted workload on a storage volume based on I/O patterns for All Flash systems. This helped in optimizing data storage and management while keeping the nature of data private to the users.
- *File and Volume Clones (FVCs)*: Developed features and managed issues related to file and volume clones on UNIX-based NetApp proprietary WAFL file system to improve the storage efficiency of enterprise data servers.
- Co-developed a file system component which prevented data loss if FVCs are modified when cloud backup is offline.

### Tata Consultancy Services Ltd.

#### iOS Application Developer

Pune, India

Mar. 2012 - Aug. 2013

- Built interactive applications for iPhone and iPad and developed tools to analyze the user investment portfolio.
- Performed cross-platform application development of mobile applications supported on both Android and iOS.

## Teaching Experience

---

### CSE 571 Artificial Intelligence [Graduate-level Course]

Arizona State University

Tempe, USA

Spring'22, Fall'19

- Delivered hands-on sessions on Planning Graphs, Markov Decision Processes, Reinforcement Learning, and Neural Networks in Spring 2022. *[Won ASU GPSA Teaching Excellence Award in Spring 2022 for this]*.
- Delivered tutorials and readings on Robot Operating System (ROS), Constraint Satisfaction Problems (CSPs), First Order Logic, Planning Domain Definition Language (PDDL), and Dynamic Bayesian Networks in Fall 2019.
- Co-created the software test-bed used to support the ROS-based course projects.
- Created and graded homework assignments and exams.
- Moderated research paper discussions in lectures.
- Mentored project groups working on course projects.

### Session Lead [Graduate-level Courses]

Arizona State University

Tempe, USA

2020-2022

- Delivered tutorials on *Reinforcement Learning* and *Neural Networks* in CSE 471 Intro. to AI, Fall'22 (85 students).
- Moderated three research paper discussions in CSE 574 Planning and Learning Methods in AI, Spring'21 (39 students).
- Delivered tutorial on *Probabilistic Inference* in CSE 471 Intro. to AI, Fall'20 (118 students).

### CS 566 Speech Processing [Graduate-level Course]

Indian Institute of Technology Guwahati

Guwahati, India

Fall 2014

- Co-developed a custom C++ API to use a voice recording tool that students used in course projects.
- Mentored students working on course projects and homework assignments.
- Graded homework assignments and exams.

### Undergraduate Lab. Courses

Indian Institute of Technology Guwahati

Guwahati, India

2013-2015

- TA for the lab courses: CS 513 Programming Lab.; CS 462 Graphics Lab.; and CS 243 Software Engineering Lab.
- Mentored project groups working on course projects.
- Graded homework assignments and exams.

## Service

---

|              |                                                                                                                                                                                                                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Co-organizer | IJCAI-ECAI 2022 Workshop on Generalization in Planning (GenPlan'22) <a href="#">↗</a>                                                                                                                                                                                                                                   |
| PC Member    | IJCAI 2023 <a href="#">↗</a> , AAMAS 2023 <a href="#">↗</a> , AAAI 2023 <a href="#">↗</a> , ICAPS 2022 <a href="#">↗</a> , XAIP'22 <a href="#">↗</a> , XAIP'21 <a href="#">↗</a> , GenPlan'21 <a href="#">↗</a>                                                                                                         |
| Reviewer     | IEEE Robotics and Automation Letters <a href="#">↗</a> , Graduate and Professional Student Association, ASU <a href="#">↗</a>                                                                                                                                                                                           |
| Volunteer    | IJCAI-ECAI 2022 <a href="#">↗</a> , KR 2022 <a href="#">↗</a> , IJCAI 2021 <a href="#">↗</a> , ICAPS 2021 <a href="#">↗</a> , ICLR 2021 <a href="#">↗</a> , KR 2018 <a href="#">↗</a> , Southwest Robotics Symposium 2019 <a href="#">↗</a> , Workshop on GPU Programming and Applications 2014, IITG <a href="#">↗</a> |

## Honors & Awards

---

|                     |                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------|
| July 2022           | <b>IJCAI-AIJ 2022 Travel and Accessibility Grant</b> , IJCAI-ECAI 2022                              |
| July 2022           | <b>KR 2022 Student Grant</b> , KR 2022                                                              |
| May 2022            | <b>Best Demo Award</b> , AAMAS 2022                                                                 |
| May 2022            | <b>ICAPS Student Scholarship</b> , ICAPS 2022                                                       |
| April 2022          | <b>GPSA Teaching Excellence Award</b> , Arizona State University                                    |
| March 2022          | <b>SCAI Doctoral Fellowship</b> , School of Computing and AI, Arizona State University              |
| 2021-2022           | <b>GPSA Travel Grant</b> , Arizona State University (for AAAI 2021, AAAI 2022, IJCAI 2022, KR 2022) |
| Mar 2021            | <b>Graduate College Travel Awards</b> , ASU (for AAMAS 2021, ICLR 2021, ICML 2021, IJCAI 2021)      |
| Oct 2018            | <b>Fulton Schools Experiential Learning Grant</b> , Arizona State University                        |
| Aug 2013 - May 2015 | <b>Post Graduation Fellowship</b> , All India Council for Technical Education, Government of India  |
| June 2012           | <b>TCS ILP Kudos Award</b> , Tata Consultancy Services Ltd. (Best Trainee in the new-joiner batch)  |

## Mentorship

---

### Trevor Angle

Tempe, USA

M.S. in Computer Science, Arizona State University

2021-2022

- Co-mentored Trevor at ASU where he worked on a system for skill-aligned explainable robot planning, called JEDAI.
- Resulted in one short paper published in demonstration track at AAMAS 2022.

### Shashank Rao Marpally

Tempe, USA

M.S. in Robotics and Autonomous Systems, Arizona State University

2020-2021

- Mentored Shashank for 3 semesters during his masters where he worked on learning action models for game agents.
- Resulted in two conference publications (AAAI'21 and KR'22).

### Masih Tamsoy

Guwahati, India

B.Tech. in Computer Science and Engineering, IIT Guwahati

2014-2015

- Mentored Masih for 2 semesters at Robotics Lab, IIT Guwahati where he worked on his bachelor's thesis titled "Speaker Verification Systems".
- Helped Masih in developing a speaker verification system using joint factor analysis and i-vectors.

## Other Affiliations

---

|                      |                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------|
| Jan 2021 - (present) | <b>Graduate Student</b> , Center for Human, Artificial Intelligence, and Robot Teaming, ASU <a href="#">↗</a>   |
| Nov 2020 - (present) | <b>Affiliated Graduate Student</b> , Center for Human-Compatible Artificial Intelligence, UCB <a href="#">↗</a> |
| Nov 2020 - (present) | <b>Member</b> , Association for the Advancement of Artificial Intelligence (AAAI) <a href="#">↗</a>             |
| July 2019            | <b>Participant</b> , 3rd Summer School on Cognitive Robotics, USC <a href="#">↗</a>                             |
| Aug 2018 - (present) | <b>Graduate Student</b> , Autonomous Agents and Intelligent Robots (AAIR) Lab, ASU <a href="#">↗</a>            |
| Jul 2016 - Dec 2016  | <b>Online Contributor</b> , Stanford Scholar Initiative <a href="#">↗</a>                                       |
| Jan 2014 - May 2015  | <b>Graduate Student</b> , Robotics Lab, IIT Guwahati <a href="#">↗</a>                                          |

---